(a) A longest induced path from s



Each extra neighbour y sees s or b (dashed edges).
The edge ay would create a 4-cycle.

(b) Deleting two degree-2 endpoints



If $d(b) = d(c) = 2$, delete $U = \{b, c\}$.
The path a, s, y survives and $D_G(U) = 6$.